

GUOCONG ZHAI

School of Transportation and Logistics, Southwest Jiaotong University
Chengdu, Sichuan, China — gzhai@swjtu.edu.cn

EDUCATION

Ph.D. in Transportation Engineering

January 2021–May 2024

Old Dominion University, Norfolk, Virginia, USA

Dissertation: Advancing Data-Driven Decision Making for Transportation Safety: Emerging Trends, Statistical Modeling, and Causal Inference

Advisor: Associate Professor Kun Xie

M.S. in Transportation Planning and Management

September 2017–June 2020

Southwest Jiaotong University, Chengdu, Sichuan, China

Thesis: Ride-hailing User Responses to Service Suspension: A Stated Preference Analysis

Advisor: Professor Hongtai Yang

B.S. in Transportation Safety Engineering

September 2012–June 2016

Chang'an University, Xi'an, Shaanxi, China

Thesis: Voice Interventions and Speeding Behavior Across Driver Psychological Profiles

Advisor: Professor Shifeng Niu

APPOINTMENTS

Assistant Professor

May 2025–Present

School of Transportation and Logistics, Southwest Jiaotong University, China

Research Fellow

September 2024–July 2025

Department of Civil and Environmental Engineering, National University of Singapore, Singapore

Mentor: Assistant Professor Prateek Bansal

Graduate Research Assistant

January 2021–August 2024

Department of Civil and Environmental Engineering, Old Dominion University, USA

Advisor: Associate Professor Kun Xie

Graduate School Assistant

August 2016–June 2017

Graduate School, Chang'an University, China

SELECTED RESEARCH FUNDING

Causal Decision-Making for Safe Systems

2027–2029

Ministry of Education of the People's Republic of China, Overseas Postdoctoral Talent Program

Principal Investigator

¥930,000 CNY

Multimodal AI-Driven Demand Identification and Route Planning for Urban Air Mobility

2026–2027

Fundamental Research Funds for the Central Universities

Co-Principal Investigator, 2 of 2

¥100,000 CNY

Digital Sisters: A Causally-Grounded Framework for Urban Mobility Knowledge Transfer in Data-Scarce Cities

2026–2027

National Natural Science Foundation of China

Co-Principal Investigator, 2 of 3

¥200,000 CNY

Causal Inference Approach to Missing-Not-at-Random Mechanism Identification and Collaborative Imputation in Crash Data

2026–2028

National Natural Science Foundation of China, Young Scientists Fund

Principal Investigator

¥300,000 CNY

Home and Firm Location Choice Models and Platform Development

2024–2027

Urban Redevelopment Authority, Singapore

Research Fellow

Factors Influencing Pedestrian Decisions to Cross Mid-Block and Potential Countermeasures

2022–2024

Federal Highway Administration and Virginia Department of Transportation

Research Assistant

Improving Safety Service Patrol Performance

2021–2023

Federal Highway Administration and Virginia Department of Transportation

Research Assistant

Statewide Landslide Risk Assessment on Transportation Infrastructure in Virginia 2022–2024

Old Dominion University Internal Research Grant

Research Assistant

PUBLICATIONS

Theme I: Causal Inference for Transportation Safety

- [S1] Pu, Q., Xie, K., Yang, H., & Zhai, G. (2026). A vision-and-knowledge enhanced large language model for generalizable pedestrian crossing behavior inference. *under review at Travel Behaviour and Society*.
- [S2] Pu, Q., Xie, K., Zhu, Y., & Zhai, G. (2026). Generating realistic safety-critical scenarios for vehicle-pedestrian interactions. *under review at Accident Analysis and Prevention*.
- [S3] Zhai, G., Xie, K., Yang, D., & Yang, H. (2025). Do electric vehicles lead to more severe crashes? A doubly robust-based causal inference approach. *Under review at Transportation Research Part A: Policy and Practice*.
- [S4] Zhai, G., Xie, K., Yang, D., & Yang, H. (2024). Developing equity-aware safety performance functions for identifying hotspots of pedestrian-involved crashes. *Accident Analysis & Prevention*, 207, 107759.
- [S5] Zhai, G., Xie, K., Yang, H., & Yang, D. (2023). Are ride-hailing services safer than taxis? A multivariate spatial approach with accommodation of exposure uncertainty. *Accident Analysis & Prevention*, 191, 107201.
- [S6] Zhai, G., Xie, K., Yang, D., & Yang, H. (2022). Assessing the safety effectiveness of citywide speed limit reduction: A causal inference approach integrating propensity score matching and spatial difference-in-differences. *Transportation Research Part A: Policy and Practice*, 157, 94–106.
- [S7] Wang, J., Cetin, M., Yang, H., Xie, K., Zhai, G., & Ishak, S. (2024). Improving effectiveness of the safety service patrol programs: A discrete event-based simulation approach. *Transportation Research Record*, 2678(8), 1–14.
- [S8] Wu, Z., Zhai, G., & Bansal, P. (2026). Causal inference in conjoint analysis: Logit models vs. potential outcomes. *Journal of Choice Modelling* (forthcoming).
- [S9] Zhai, G., & Yang, H. (2020). Is the front passenger seat always the “death seat”? An application of a hierarchical ordered probit model for occupant injury severity. *International Journal of Injury Control and Safety Promotion*, 27(4), 438–446.

Theme II: AI-Driven Behavioral Analytics for Sustainable Shared Mobility

- [M1] Zhai, G., Yang, H., Yang, Z., Chen, S., Jiang, L., & Khattak, A. J. (2026). Assessing the effects of congestion surcharge on taxi ridership: Disentangling spatial spillover, nonlinearity, and time trends. *Under review at Transport Policy*.
- [M2] Liu, X., Zhai, G., Yang, H., Wang, R., Wang, D., & Witlox, F. (2026). Modeling taxi-metro competition: Uncovering nonlinear influences of trip and built environment factors. *Under review at Transportation Research Part D: Transport and Environment*.
- [M3] Zhang, H., Xue, X., Zhai, G., Yang, H., Lu, X., & Cherry, C. (2026). Choice preferences for urban air mobility: A latent class discrete choice model with mixed logit extensions. *Travel Behaviour and Society*, 43, 101229.
- [M4] Yang, Z., Zhai, G., Sze, N. N., Ding, H., Bobylev, N., & Yang, H. (2026). Exploring different patterns of bike-and-ride trips and influencing factors using geographically weighted random forest. *Travel Behaviour and Society*.
- [M5] Zhai, G., Wang, R., Liu, X., Mladenović, M. N., Tang, Y., Mu, H., Liu, X., & Yang, H. (2025). Built environment impacts on zonal shared e-scooter expenses: A Bayesian learning approach. *Transportation Research Part D: Transport and Environment*, 148, 105020.
- [M6] Yang, H., Luo, P., Li, C., & Zhai, G. (2023). Nonlinear effects of fare discounts and built environment on ridesplitting adoption rates. *Transportation Research Part A: Policy and Practice*, 168, 103–118.
- [M7] Yang, H., An, P., Wei, H., & Zhai, G. (2023). Unique in the metro system: The likelihood to re-identify a metro user with limited trajectory points. *Physica A: Statistical Mechanics and Its Applications*, 610, 128–141.
- [M8] Yang, H., Zhai, G., & Yang, L. (2022). How does the suspension of ride-sourcing affect the transportation system and environment? *Transportation Research Part D: Transport and Environment*, 102, 103131.
- [M9] Zhang, Z., Zhai, G., Xie, K., & Xiao, F. (2022). Exploring the nonlinear effects of ridesharing on public transit usage: A case study of San Diego. *Journal of Transport Geography*, 104, 103449.
- [M10] Yang, H., Guo, Z., Zhai, G., Yang, L., Huo, J., & Meng, X. (2022). Exploring the spatially heterogeneous effects of the built environment on bike-sharing usage during the COVID-19 pandemic. *Journal of Advanced Transportation*, 2022, 287–301.
- [M11] Yang, H., Zhai, G., Liu, X., Yang, L., Liu, Y., & Yuan, Q. (2022). Determinants of city-level private car ownership: Effects of vehicle regulation policies and relative prices. *Transport Policy*, 115, 40–48.

PROFESSIONAL SERVICE

Editorial Service

- Guest Editor, *Transportation Research Part D: Transport and Environment* (2025). Organized a special issue on *AI-Driven Behavioral Analytics for Sustainable Shared Mobility: Advancing Models, Policies, and Implementation*.
- Youth Editorial Board Member, *Digital Transportation and Safety* (2024–Present).
- Youth Editorial Board Member, *Transportation Safety and Environment* (2023–Present).

Peer Review

Completed more than 60 peer reviews for journals including *Transportation Research Parts A–D*, *Safety Science*, *Accident Analysis & Prevention*, *Journal of Transport Geography*, *Transportation*, *Transport Policy*, and *Cities*.

Outstanding Reviewer, *Transportation Research Part D*, 2021.

AWARDS AND HONORS

- Academic Research Excellence Award, Old Dominion University College of Engineering (2023). Sole recipient school-wide.
- Rollie Dubbe Engineering Scholarship, Old Dominion University (2022). Sole recipient university-wide.
- Outstanding Reviewer Award, *Transportation Research Part D* (2021).
- Outstanding Graduate, Southwest Jiaotong University (2020).

METHODS AND TECHNICAL SKILLS

- **Causal inference & statistical learning:** propensity score methods, doubly robust estimation, Bayesian machine learning
- **Generative AI & LLM toolkits:** large language model prompting and fine-tuning, retrieval-augmented generation (RAG), vision-language models
- **Programming:** Python (Pandas, NumPy, Scikit-learn, Statsmodels, GeoPandas, PyTorch, Transformers), R, SQL, and MATLAB
- **Geographic information systems:** ArcGIS and QGIS